

**CRIMINAL ACTIVITY NUISANCE ORDINANCE  
(CANO) DATABASE**  
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# CRIMINAL ACTIVITY NUISANCE ORDINANCES (CANOs)

- ▶ **What they are:** municipal ordinances that treat *repeated “nuisance” incidents* as a *property-level problem*, triggering enforcement against owners/landlords and occupants.
- ▶ **Why they matter:** prior case-study evidence links CANOs to increased eviction risk and raises procedural concerns.
- ▶ **The gap:** despite widespread adoption, there is **no comprehensive national database** of where CANOs exist or how they are designed.
- ▶ **This chapter:** builds the first national CANO database from scraped municipal codes and uses it to study prevalence, correlates, and impacts on evictions and crime.

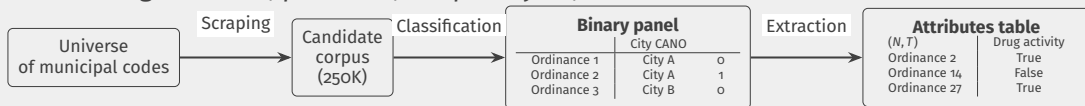
▶ Definition details

# DATA AND CLASSIFICATION PIPELINE

- **Data collection:** Scrape and parse thousands of municipal codes across the U.S.
- **Classification:** Use a Transformer classifier (LED) to classify CANOs.

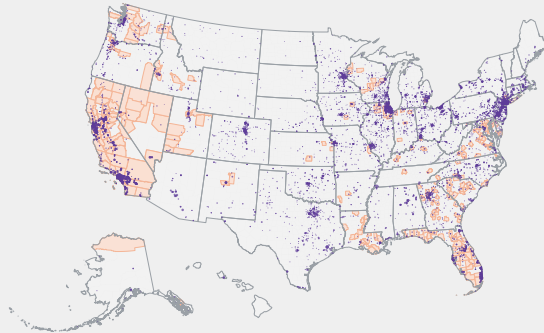
| Model                            | N     | TP | TN  | FP  | FN | Accuracy     | Recall       | FPR          | F1           |
|----------------------------------|-------|----|-----|-----|----|--------------|--------------|--------------|--------------|
| 40-mini                          | 1,000 | 40 | 753 | 203 | 4  | 0.793        | 0.909        | 0.212        | 0.279        |
| 40-mini fine-tuned               | 1,000 | 34 | 915 | 41  | 10 | 0.949        | 0.773        | 0.043        | 0.571        |
| 40-mini fine-tuned (balanced)    | 1,000 | 32 | 862 | 94  | 12 | 0.894        | 0.727        | 0.098        | 0.376        |
| <b>LED (best threshold)</b>      | 1,000 | 41 | 933 | 23  | 3  | <b>0.974</b> | <b>0.932</b> | <b>0.024</b> | <b>0.759</b> |
| LED (0.5 threshold)              | 1,000 | 41 | 906 | 50  | 3  | 0.947        | 0.932        | 0.052        | 0.607        |
| 40-mini and LED (best threshold) | 1,000 | 39 | 940 | 16  | 5  | 0.979        | 0.886        | 0.017        | 0.788        |
| 40-mini and LED (0.5 threshold)  | 1,000 | 39 | 928 | 28  | 5  | 0.967        | 0.886        | 0.029        | 0.703        |

- **Attribute extraction:** Extract policy features using LLMs (gpt-40-mini) such as warning structure, penalties, adoption year, etc.



# DISTRIBUTION OF CANOS

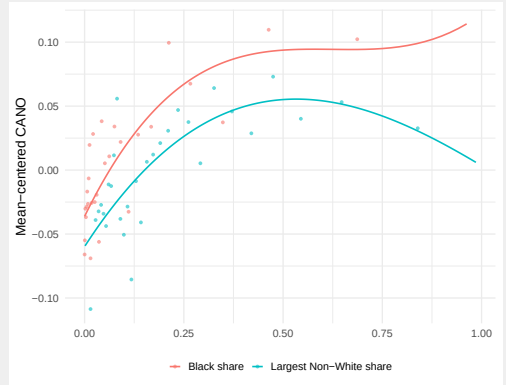
- We identify 8,000 ordinances that link to their jurisdictions, and link jurisdictions with census covariates, panels of eviction and crime data.



Geographic distribution of predicted CANOs (according to the LED best threshold) in incorporated places (dots) and counties (shaded regions).

# RACIAL THREAT AND CANO ADOPTION

- ▶ Following Moy (2024), interpret CANO adoption through a classic **racial threat** lens: punitive tools become more attractive as minority presence grows.
- ▶ Nationally, jurisdictions with larger Black or non-White population shares are substantially more likely to have a CANO, even after rich demographic and socioeconomic controls.
- ▶ The relationship is non-monotone: CANO incidence peaks at intermediate minority shares.

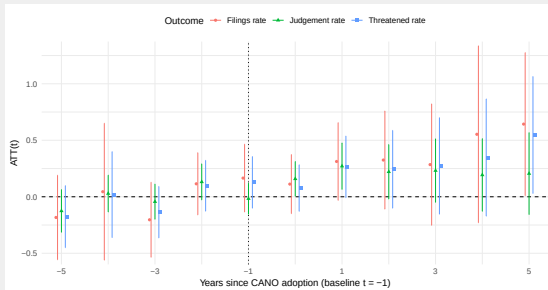


▶ Racial threat regressions 1

▶ Racial threat regressions 2

# CANOS AND OUTCOMES

- **External evidence:** Kroeger and La Mattina (2020) show that Ohio cities adopting CANOs experience about 0.4 additional evictions and roughly 0.9 additional filings per 100 renter homes.
- **Our results:** Using national eviction panels, CANO adoption is associated with an increase of about 0.25 eviction filings and 0.4 evictions per 100 renter homes.



THANK YOU!

# REFERENCES I

Moy, B. J. (2024). Racial threat and the emergence of discriminatory ordinances.

Kroeger, S., & La Mattina, G. (2020). Do nuisance ordinances increase eviction risk? *AEA Papers and Proceedings*, 110, 452–56. <https://doi.org/10.1257/pandp.20201119>



# APPENDIX

# WHAT IS A CANO? (OPERATIONAL DEFINITION)

- ▶ A CANO typically has four components:
  - 1 **Definition of nuisance activities:** a list of incidents that “count” as nuisance.
  - 2 **Nuisance formula / threshold:** a rule like “ $N$  incidents within  $T$  months”.
  - 3 **Penalties:** fines/cost recovery/other sanctions once a property is designated.
  - 4 **Abatement duties:** actions required (or pressured) to prevent recurrence; can include eviction-related provisions.
- ▶ A similar class of ordinances *Crime-free housing ordinances* (CFHOs): local laws that require landlords to evict tenants for alleged criminal activity or nuisance behavior, often through mandatory “crime-free lease addendums” and police involvement.

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**Table:** Summary of article counts by jurisdiction.

| Region | N     | Articles |        |         |     |        |
|--------|-------|----------|--------|---------|-----|--------|
|        |       | Mean     | Median | SD      | Min | Max    |
| City   | 7,642 | 29.2     | 22     | 26.7    | 1   | 392    |
| County | 578   | 34.1     | 25     | 30.1    | 1   | 251    |
| State  | 50    | 4,640.7  | 2,117  | 5,420.4 | 113 | 26,301 |

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Table 2: CANO vs. Black race and controls.

|                            | (1)                 | (2)                  | (3)                 | (4)                 |
|----------------------------|---------------------|----------------------|---------------------|---------------------|
| Black share                | -0.100**<br>(0.046) | -0.490***<br>(0.112) | 0.202***<br>(0.051) | 0.403***<br>(0.123) |
| Black share <sup>2</sup>   |                     | 0.674***<br>(0.176)  |                     | -0.318*<br>(0.178)  |
| Log population             | 0.005<br>(0.004)    | 0.009**<br>(0.004)   | 0.035***<br>(0.005) | 0.033***<br>(0.005) |
| Poverty rate               | -0.280**<br>(0.113) | -0.291***<br>(0.113) | -0.140<br>(0.108)   | -0.131<br>(0.109)   |
| Renter share               | 0.495***<br>(0.057) | 0.497***<br>(0.057)  | -0.010<br>(0.062)   | -0.018<br>(0.062)   |
| Crime share                | 0.002<br>(0.003)    | 0.003<br>(0.003)     | 0.000<br>(0.003)    | 0.000<br>(0.003)    |
| Eviction filings share     | -0.152<br>(0.835)   | -0.118<br>(0.834)    | 0.427<br>(0.857)    | 0.447<br>(0.857)    |
| Eviction judgements share  | 6.156**<br>(2.531)  | 6.123**<br>(2.529)   | 7.965***<br>(2.594) | 8.042***<br>(2.593) |
| Num.Obs.                   | 6345                | 6345                 | 6345                | 6345                |
| R2                         | 0.017               | 0.019                | 0.175               | 0.176               |
| R2 Adj.                    | 0.016               | 0.018                | 0.163               | 0.164               |
| State-Jurisdiction type FE |                     |                      | ✓                   | ✓                   |

\* p <0.1, \*\* p <0.05, \*\*\* p <0.01

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Table 3: CANO vs. largest non-White share and controls.

|                                      | (1)                  | (2)                  | (3)                 | (4)                 |
|--------------------------------------|----------------------|----------------------|---------------------|---------------------|
| Largest Non-White share              | 0.119***<br>(0.034)  | 0.154<br>(0.099)     | 0.062*<br>(0.037)   | 0.307***<br>(0.104) |
| Largest Non-White share <sup>2</sup> |                      | -0.047<br>(0.124)    |                     | -0.308**<br>(0.123) |
| Log population                       | 0.001<br>(0.004)     | 0.001<br>(0.004)     | 0.037***<br>(0.005) | 0.035***<br>(0.005) |
| Poverty rate                         | -0.440***<br>(0.113) | -0.439***<br>(0.113) | -0.096<br>(0.109)   | -0.083<br>(0.109)   |
| Renter share                         | 0.475***<br>(0.057)  | 0.473***<br>(0.057)  | -0.000<br>(0.062)   | -0.018<br>(0.062)   |
| Crime share                          | 0.003<br>(0.003)     | 0.003<br>(0.003)     | 0.000<br>(0.003)    | 0.000<br>(0.003)    |
| Eviction filings share               | -0.979<br>(0.808)    | -0.996<br>(0.809)    | 0.993<br>(0.844)    | 0.917<br>(0.844)    |
| Eviction judgements share            | 6.412**<br>(2.531)   | 6.418**<br>(2.531)   | 8.519***<br>(2.593) | 8.393***<br>(2.592) |
| Num.Obs.                             | 6345                 | 6345                 | 6345                | 6345                |
| R2                                   | 0.018                | 0.018                | 0.174               | 0.174               |
| R2 Adj.                              | 0.017                | 0.017                | 0.162               | 0.162               |
| State-Jurisdiction type FE           |                      |                      | ✓                   | ✓                   |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01

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| Group                        | Estimate (SE)   | N    |
|------------------------------|-----------------|------|
| <b>Filings rate</b>          |                 |      |
| All                          | 0.425** (0.212) | 1022 |
| Eviction mandated            | 0.490* (0.295)  | 447  |
| Eviction silent              | -0.068 (0.475)  | 166  |
| Eviction strongly encouraged | 0.274 (0.267)   | 359  |
| <b>Judgement rate</b>        |                 |      |
| All                          | 0.225* (0.132)  | 1022 |
| Eviction mandated            | 0.211 (0.205)   | 447  |
| Eviction silent              | 0.044 (0.283)   | 166  |
| Eviction strongly encouraged | 0.131 (0.167)   | 359  |
| <b>Threatened rate</b>       |                 |      |
| All                          | 0.332* (0.171)  | 1022 |
| Eviction mandated            | 0.360 (0.238)   | 447  |
| Eviction silent              | -0.145 (0.374)  | 166  |
| Eviction strongly encouraged | 0.156 (0.202)   | 359  |

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**Table:** Static DiD estimates of CANO effects on arrest rates.

| Outcome                            | Estimate (SE)  | N   |
|------------------------------------|----------------|-----|
| Drug arrests rate                  | -0.891 (0.835) | 929 |
| Disorderly arrests rate            | -0.632 (0.614) | 929 |
| Drug arrests (juvenile) rate       | -0.822 (0.915) | 929 |
| Disorderly arrests (juvenile) rate | 0.034 (0.064)  | 929 |
| Drug arrests (Black) rate          | -0.125 (0.184) | 929 |
| Disorderly arrests (Black) rate    | -0.039 (0.096) | 929 |
| Drug arrests (Hispanic) rate       | 0.074 (0.052)  | 929 |
| Disorderly arrests (Hispanic) rate | 0.022 (0.019)  | 929 |